CS60050 - Machine Learning

Assignment 3

Country Grouping using Complete Linkage Divisive (Top-Down) Clustering Technique

Bratin Mondal

21CS10016

Department of Computer Science and Engineering,
Indian Institute of Technology Kharagpur

1 Problem Statement

- Implement K-means clustering algorithm from scratch for $k = 3, 4, 5, 6$.
- Find the optimal value of k using the silhouette coefficient metric.
- Implement Complete Linkage Divisive (Top-Down) Clustering Technique from scratch for the optimal value of k found in previous step.
- Establish a one-to-one correspondence between the clusters obtained from K-means and Complete Linkage Divisive (Top-Down) Clustering Technique and calculate the Jaccard Similarity.

2 Model Description

- **K-means Clustering Algorithm:** K-means clustering is a type of unsupervised learning, which is used when you have unlabeled data (i.e., data without defined categories or groups). The goal of this algorithm is to find groups in the data, with the number of groups represented by the variable K. The algorithm works iteratively to assign each data point to one of K groups based on the features that are provided. Data points are clustered based on feature similarity.
- **Silhouette Coefficient:** The silhouette value is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette ranges from -1 to 1, where a high value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters. If most objects have a high value, then the clustering configuration is appropriate. If many points have a low or negative value, then the clustering configuration may have too many or too few clusters.
- **Complete Linkage Divisive (Top-Down) Clustering Technique:** The complete linkage method is a hierarchical clustering method that uses the maximum distance between clusters to determine the distance between clusters. In this method, the distance between two clusters is the maximum distance between their individual components. The divisive clustering technique is a top-down approach where all the data points are considered as a single cluster and then the cluster is divided into smaller clusters until each data point is a separate cluster.
- **Jaccard Similarity:** The Jaccard similarity coefficient of two sets is defined as the size of the intersection divided by the size of the union of the two sets. It is a measure of the similarity between two sets.

3 Results

3.1 K-means Clustering

The optimal value of k was found to be 3 using the silhouette coefficient metric. We observe that the silhouette

Value of K	Silhouette Coefficient
3	0.6988985969993166
4	0.6137622384616047
5	0.5641146777848881
6	0.4879572689267768

Table 1: Silhouette Coefficients for Different Values of K

coefficient decreases as the value of k increases. Hence, the optimal value of k is 3. Analysing the change of silhouette coefficient with respect to k is shown below

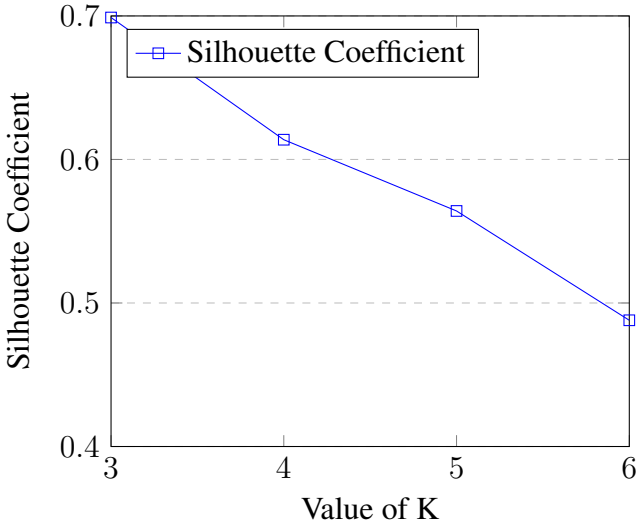

Figure 1: Silhouette Coefficients for Different Values of K

The decrease in silhouette coefficient with increasing k suggests that the clusters become less distinct as more clusters are introduced. This indicates that the data may not naturally divide into more than three clusters, leading to a decrease in the quality of clustering as k increases.

3.2 Complete Linkage Divisive (Top-Down) Clustering Technique

The Silhouette Coefficient for the optimal value of k was found to be 0.6895572393550293.

This high Silhouette Coefficient indicates that the clusters formed by the Complete Linkage Divisive clustering technique are well-separated and dense, with each data point closely resembling its own cluster centroid compared to others. It suggests that the chosen value of k effectively captures the underlying structure of the data. However, further analysis and validation may be necessary to ensure the robustness and generalizability of the clustering results.

3.3 Jaccard Similarity

The Jaccard Similarity between the clusters obtained from K-means and Complete Linkage Divisive (Top-Down) Clustering Technique is shown in the Table below.

K-Means Cluster Index	Divisive Cluster Index	Jaccard Similarity
1	1	0.8787878787878788
2	3	0.8315789473684211
3	2	0.6610169491525424

Table 2: Comparison of K-Means and Divisive Clustering Indices with Jaccard Similarity

The Jaccard Similarity measures the similarity between two sets by comparing their intersection to their union. In the context of clustering, it quantifies the similarity of cluster assignments between different clustering techniques.

From Table 2, we observe that the Jaccard Similarity values range between 0.66 and 0.88, indicating moderate to high similarity between the cluster assignments produced by K-means and Complete Linkage Divisive clustering. Higher Jaccard Similarity values suggest a greater agreement between the clustering results of the two techniques.

However, it's essential to note that while Jaccard Similarity provides valuable insights into the agreement between clustering results, it does not account for the structure or quality of the clusters themselves. Therefore, further analysis, such as examining the cluster centroids or silhouette scores, may be necessary to fully assess the clustering performance.

4 Analysis

Table 3 shows the time taken for each step of the process. The "Load Data" step, which involves loading the dataset, took a minimal amount of time, while steps such as "Find Best K" and "Hierarchical Clustering" required more computation.

The total time taken for all steps was **2.0712053775787354** seconds.

This breakdown of time taken for each step provides valuable insights into the computational overhead of the process and can help identify potential bottlenecks or areas for optimization.

Step	Time Taken (seconds)
Load Data	0.0031189918518066406
Clustering with K = 3	0.26228976249694824
Clustering with K = 4	0.24265837669372559
Clustering with K = 5	0.25972580909729004
Clustering with K = 6	0.27604103088378906
Find Best K	1.0408053398132324
Hierarchical Clustering	1.0254216194152832
Calculate Jaccard Similarity	0.0003230571746826172
Total Time Taken	**2.0712053775787354**

Table 3: Time Taken for Each Step

5 Conclusion

In this study, we implemented the K-means clustering algorithm from scratch for $k = 3, 4, 5, 6$ and found the optimal value of k to be 3 using the silhouette coefficient metric. We also implemented the Complete Linkage Divisive (Top-Down) Clustering Technique for the optimal value of k and obtained a high Silhouette Coefficient, indicating well-separated and dense clusters.

Additionally, we established a one-to-one correspondence between the clusters obtained from K-means and Complete Linkage Divisive clustering and calculated the Jaccard Similarity. The moderate to high Jaccard Similarity values suggested a reasonable agreement between the cluster assignments produced by the two techniques.

Furthermore, the breakdown of time taken for each step provided valuable insights into the computational overhead of the process. Steps such as finding the best k and performing hierarchical clustering contributed significantly to the total computation time.

Overall, this analysis sheds light on the effectiveness and efficiency of different clustering techniques in partitioning data and provides a foundation for further exploration and optimization in future studies.